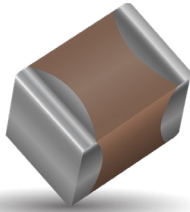


Automotive MLCC, KAM Series

General Specifications



GENERAL DESCRIPTION

KYOCERA AVX has supported the Automotive Industry requirements for Multilayer Ceramic Capacitors consistently for more than 25 years. Products have been developed and tested specifically for automotive applications and all manufacturing facilities are QS9000 and VDA 6.4 approved.

KYOCERA AVX is using AECQ200 as the qualification vehicle for this transition. A detailed qualification package is available on request and contains results on a range of part numbers.

The KAM series are plated with a Nickel/Tin finish. For FLEXITERM® please refer to the KAF series datasheet.

HOW TO ORDER

KAM	31	G	R7	1H	475	K	U
Series AEC-Q200 Tin Nickel Finish	Size 03 = 0201 05 = 0402 15 = 0603 21 = 0805 31 = 1206 32 = 1210 42 = 1808 43 = 1812 55 = 2220	Thickness See Cap Chart	Dielectric CG = COG R7 = X7R S7 = X7S T7 = X7T R8 = X8R L8 = X8L G8 = X8G	Voltage 0G = 4V 0J = 6.3V 1A = 10V 1C = 16V 1E = 25V 1H = 50V 2A = 100V 2D = 200V 2E = 250V 2H = 500V 2J = 630V 3A = 1000V 3N = 1500V 3D = 2000V 3E = 2500V 3U = 3000V	Capacitance Code Code (in pF) 2 Significant Digits +Number of zeros eg 10uF = 106 10nF = 103 47pF = 470	Capacitance Tolerance B = $\pm 0.1\text{pF}$ (<10pF)* C = $\pm 0.25\text{pF}$ (<10pF)* D = $\pm 0.5\text{pF}$ (<10pF)* F = $\pm 1\%$ * G = $\pm 2\%$ * J = $\pm 5\%$ K = $\pm 10\%$ M = $\pm 20\%$ *COG only	Packaging See Table Below

PACKAGING CODES

Code	EIA (inch)	IEC(mm)	7" Paper	7" Embossed	13" Paper	13" Embossed
03	0201	0603	H		N	
05	0402	1005	H		N	
15	0603	1608	T	U	M	L
21	0805	2012	T	U	M	L
31	1206	3216	T	U	M	L
32	1210	3225	T	U	M	L
42	1808	4520		Y		K
43	1812	4532		V		S
55	2220	5750		V		S

*thickness determines paper or plastic embossed packaging

DIELECTRIC

Dielectric	Operating Temperature (°C)	Capacitance Change Rate
X7R	-55~+125	$\pm 15\%$
X7T	-55~+125	$\pm 22/-33\%$
X8R	-55~+150	$\pm 15\%$
X8L	-55~+125	$\pm 15\%$
X8L	+125~+150	+15/-40%
X8G	-55~+150	0 $\pm 30\text{ppm}/^\circ\text{C}$
NP0	-55~+125	0 $\pm 30\text{ppm}/^\circ\text{C}$

TYPICAL APPLICATIONS

X7R KAM

- High capacitance values
- Broadest voltage and cap offering
- Cameras
- Body control modules
- Infotainment
- ECU
- Climate control

X7T KAM

- Motor drive
- Door lock

NP0 KAM

- Extreme capacitance stability
- Automotive LED Lighting System
- Key fob
- Audio
- Touchscreen
- GPS
- Safety

X8G KAM

- Extreme capacitance stability
- High temperature
- Battery Management Systems
- Powertrain Sensors & Actuators
- Engine management
- Transmission control
- Safety

Automotive MLCC, KAM Series

General Specifications



COMMERCIAL VS AUTOMOTIVE MLCC PROCESS COMPARISON

	Commercial	Automotive
Administrative	Standard Part Numbers. No restriction on who purchases these parts.	Specific Automotive Part Number. Used to control supply of product to Automotive customers.
Lot Qualification (Destructive Physical Analysis - DPA)	As per EIA RS469	Increased sample plan stricter criteria.
Visual/Cosmetic Quality	Standard process and inspection	100% inspection
Application Robustness	Standard sampling for accelerated wave solder	Increased sampling for accelerated wave solder followed by lot by lot reliability testing.

All Tests have Accept/Reject Criteria 0/1

Automotive MLCC - NP0

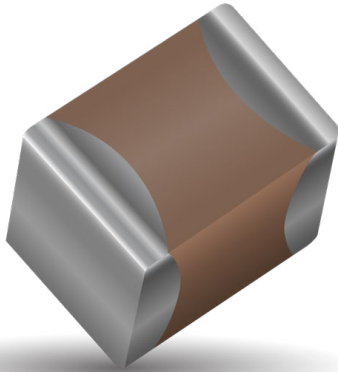
General Specifications

TYPICAL APPLICATIONS

- Extreme capacitance stability
- Automotive LED Lighting System
- Key fob
- Audio
- Touchscreen
- GPS
- Safety

ENGINEERING TOOLS

- Samples
- Technical Articles
- Application Engineering
- Application Support



Automotive MLCC - X7R / X7T

General Specifications

TYPICAL APPLICATIONS

X7R KAM

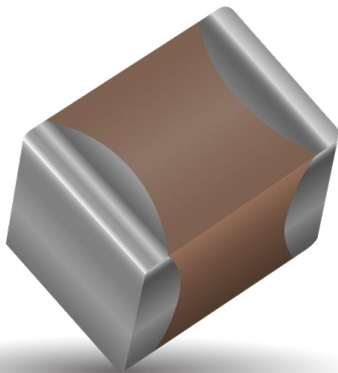
- High capacitance values
- Broadest voltage and cap offering
- Cameras
- Body control modules
- Infotainment
- ECU
- Climate control

X7T KAM

- Motor drive
- Door lock

ENGINEERING TOOLS

- Samples
- Technical Articles
- Application Engineering
- Application Support



Automotive MLCC - X7R, 4V to 500V

[illegible]

*These dimensions differ from the standard in table above and are:

Length (mm) 3.2 ± 0.4

Width (mm) 1.6 ± 0.3

Automotive MLCC - X7R, 630V to 3000V

Capacitance Range



PREFERRED SIZES ARE SHADED

Case Size	1206					1210				1808						1812						2220					
Soldering	Reflow/Wave					Reflow/Wave				Reflow Only						Reflow Only						Reflow Only					
(L) Length	mm (in.)	3.2 ± 0.2 (0.126 ± 0.008)				3.2 ± 0.2 (0.126 ± 0.008)				4.57 ± 0.25 (0.18 ± 0.01)						4.5 ± 0.3 (0.177 ± 0.012)						5.7 ± 0.5 (0.224 ± 0.02)					
(W) Width	mm (in.)	1.6 ± 0.2 (0.063 ± 0.008)				2.5 ± 0.2 (0.098 ± 0.008)				2.03 ± 0.25 (0.08 ± 0.01)						3.2 ± 0.2 (0.126 ± 0.008)						5 ± 0.4 (0.197 ± 0.016)					
(t) Terminal	mm max	0.5 ± 0.25 (0.02 ± 0.01)				0.5 ± 0.25 (0.02 ± 0.01)				0.61 ± 0.36 (0.024 ± 0.014)						0.61 ± 0.36 (0.024 ± 0.014)						0.64 ± 0.39 (0.025 ± 0.015)					
Voltage (V)		630	1000	1500	2000	2500	630	1000	1500	2000	630	1000	1500	2000	2500	3000	630	1000	1500	2000	2500	3000	630	1000	1500	2000	3000
101	Cap 100	B	B	B	B	B																					
121	(pF) 120	B	B	B	B	B																					
151	150	B	B	B	B	B																					
181	180	B	B	B	B	B																					
221	220	B	B	B	B	B					B	B	B	B	B	B											
271	270	B	B	B	B	B	H	H	H	H	B	B	B	B	B	B											
331	330	B	B	B	B	B	H	H	H	H	B	B	B	B	B	B	E										
391	390	B	B	B	B	B	H	H	H	H	B	B	B	B	B	B	E										
471	470	B	B	B	B	B	H	H	H	H	B	B	B	B	B	B	E	E	E	E	E	E					
561	560	B	B	B	B	B	H	H	H	H	B	B	B	B	B	B	E	E	E	E	E	E					
681	680	B	B	B	B	B	H	H	H	H	B	B	B	B	B	B	E	E	E	E	E	E					
821	820	B	B	B	B	B	H	H	H	H	B	B	C	C	C	C	E	E	E	E	E	E					
102	1000	B	B	B	B	B	H	H	H	H	B	B	C	C	C	C	E	E	E	E	E	E	Z	Z	Z	Z	C
122	1220	D	A	A	A		H	H	H	H							F	F	F	F	F	F	Z	Z	Z	Z	C
152	1500	D	A	A	A		H	H	H	H							F	F	F	F	F	F	Z	Z	Z	Z	C
182	1800	D	A	A			H	H	H	H							F	F	F	F	F	F	Z	Z	Z	Z	C
222	2200	D	A	A			H	H	H	H							F	F	F	F	F	F	Z	Z	Z	Z	C
272	2700	D	A	A			H	H	H	H							F	F	F	F	F	F	Z	Z	Z	Z	C
332	3300	D	A				H	H	H	H							F	F	F	F			Z	Z	Z	Z	
392	3900	D	A				H	H	H								F	F	F	F			Z	Z	Z	Z	
472	4700	D	A				H	H	H								F	F	J	J			Z	Z	Z	Z	
562	5600	D	A				H	H	H								F	F	J	J			Z	Z	Z	Z	
682	6800	A	A				H	H									F	F	J	J			Z	Z	Z	Z	
822	8200	A					H	H									F	F	J	J			Z	Z	C	C	
103	Cap 0.01	A					H	H									F	F	J				C	C	C	C	
123	(μF) 0.012						H	H									F	F	J				C	C	C	C	
153	0.015						H	H									F	F	J				C	C	C	C	
183	0.018						H										F	F	J				C	C	C	C	
223	0.022						H										F	F					C	C	C	C	
273	0.027						H										F	F					C	C	C	C	
333	0.033																F						C	C			
393	0.039																F						C	C			
473	0.047																F						C	C			
563	0.056																						C	C			
683	0.068																						C	C			
823	0.082																						C	C			
104	0.1																						C	C			
124	0.12																						C				
154	0.15																						C				
224	0.22																										
334	0.33																										
474	0.47																										
684	0.68																										
105	1																										
WVDC		630	1000	1500	2000	2500	630	1000	1500	2000	630	1000	1500	2000	2500	3000	630	1000	1500	2000	2500	3000	630	1000	1500	2000	3000
Size		1206					1210				1808						1812						2220				

NOTE: Contact factory for non-specified capacitance values

Case Size	1206(KAM31)			1210(KAM32)	1808(KAM42)		1812(KAM43)			2220(KAM55)	
Thickness Letter	B	D	A	H	B	C	E	F	J	Z	C
Max Thickness	0.94	1.45	1.80	1.80	1.80	2.21	1.80	2.21	2.80	2.21	2.80
Carrier Tape	PAPER	EMB	EMB	EMB	EMB	EMB	EMB	EMB	EMB	EMB	EMB
Packaging Code 7"reel	T	U	U	U	Y	Y	V	V	V	V	V
Packaging Code 13"reel	M	L	L	L	K	K	S	S	S	S	S
Paper						EMBOSSED (EMB)					

Automotive MLCC - X8R / X8L

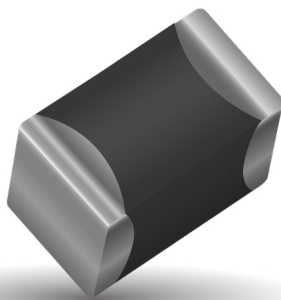
General Specifications

TYPICAL APPLICATIONS

- All market sectors with a 150°C requirement
- Automotive on engine applications
- Oil exploration applications
- Hybrid automotive applications
 - Battery control
 - Inverter / converter circuits
 - Motor control applications
 - Water pump
- Hybrid commercial applications
 - Emergency circuits
 - Sensors
 - Temperature regulation

ENGINEERING TOOLS

- Samples
- Technical Articles
- Application Engineering
- Application Support

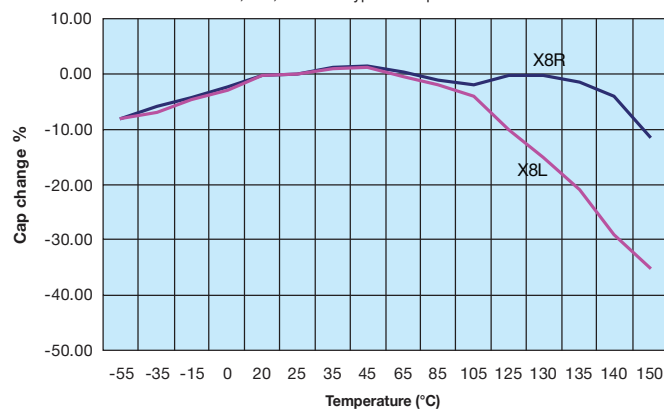


ADVANTAGES OF X8R AND X8L MLC CAPACITORS

- Both ranges are qualified to the highest automotive AEC-Q200 standards
- Excellent reliability compared to other capacitor technologies
- RoHS compliant
- Low ESR / ESL compared to other technologies
- Tin solder finish
- FLEXITERM® available
- 100V range available

X8R/X8L Dielectric

0805, 50V, X8R/X8L Typical Temperature Coefficient



Automotive MLCC - X8R / X8L

Capacitance Range

KYOCERA AVX has developed a range of multilayer ceramic capacitors designed for use in applications up to 150°C. These capacitors are manufactured with an X8R and an X8L dielectric material. X8R material has capacitance variation of $\pm 15\%$ between -55°C and +150°C. The X8L material has capacitance variation of $\pm 15\%$ between -55°C to 125°C to 125°C and +15/40% from +125°C to +150°C.

The need for X8R and X8L performance has been driven by customer requirements for parts that operate at elevated temperatures. They provide a highly reliable capacitor with low loss and stable capacitance over temperature.

They are ideal for automotive under the hood sensors, and various industrial applications. Typical industrial application would be drilling monitoring system. They can also be used as bulk capacitors for high temperature camera modules.

X8R

SIZE		0402	0603		0805		1206	
Soldering		Reflow/Wave	Reflow/Wave		Reflow/Wave		Reflow/Wave	
(L) Length	mm (in.)	1.0 $\pm$ 0.2 (0.04 $\pm$ 0.008)	1.6 $\pm$ 0.15 (0.063 $\pm$ 0.006)		2.01 $\pm$ 0.2 (0.079 $\pm$ 0.008)		3.2 $\pm$ 0.2 (0.126 $\pm$ 0.008)	
(W) Width	mm (in.)	0.5 $\pm$ 0.2 (0.02 $\pm$ 0.008)	0.81 $\pm$ 0.15 (0.032 $\pm$ 0.006)		1.25 $\pm$ 0.2 (0.049 $\pm$ 0.008)		1.6 $\pm$ 0.2 (0.063 $\pm$ 0.008)	
(t) Terminal	mm (in.)	0.25 $\pm$ 0.15 (0.01 $\pm$ 0.006)	0.35 $\pm$ 0.15 (0.014 $\pm$ 0.006)		0.5 $\pm$ 0.25 (0.02 $\pm$ 0.01)		0.5 $\pm$ 0.25 (0.02 $\pm$ 0.01)	
WVDC		50V	25V	50V	100V	25V	50V	100V
271	Cap 270	A	A	A	A			
331	(pF) 330	A	A	A	A	B	B	B
471	470	A	A	A	A	B	B	B
681	680	A	A	A	A	B	B	B
102	1000	A	A	A	A	B	B	B
152	1500	A	A	A	A	B	B	B
182	1800	A	A	A	A	B	B	B
222	2200	A	A	A	A	B	B	B
272	2700	A	A	A	A	B	B	B
332	3300	A	A	A	A	B	B	B
392	3900	A	A	A	A	B	B	B
472	4700	A	A	A	A	B	B	B
562	5600		A	A	A	B	B	B
682	6800		A	A	A	B	B	B
822	8200		A	A	A	B	B	B
103	Cap 0.01		A	A	A	B	B	B
123	(uF) 0.012		A	A		B	B	B
153	0.015		A	A		B	B	B
183	0.018		A	A		B	B	B
223	0.022		A	A		B	B	B
273	0.027		A	A		B	B	B
333	0.033		A	A		B	B	B
393	0.039		A	A		B	B	B
473	0.047		A	A		B	B	B
563	0.056		A			A	A	N
683	0.068		A			A	A	N
823	0.082					A	A	N
104	0.1					A	A	N
124	0.12					A	A	N
154	0.15					A	A	N
184	0.18					A		N
224	0.22					A		N
274	0.27							N
334	0.33							N
394	0.39							E
474	0.47							E
684	0.68							G
824	0.82							G
105	1							G
WVDC		50V	25V	50V	100V	25V	50V	100V
SIZE		0402	0603		0805		1206	
Case Size	0402(KAM05)	0603(KAM15)		0805(KAM21)		1206(KAM31)		1210(KAM32)
Thickness Letter	A	A	B	B	A	B	N	E
Max Thickness	0.56	0.90	0.95	0.94	1.45	0.94	1.27	1.52
Carrier Tape	PAPER	PAPER	PAPER	PAPER	EMB	PAPER	EMB	EMB
Packaging Code 7"reel	H	T	T	T	U	T	U	U
Packaging Code 13"reel	N	M	M	M	L	M	L	L
		Paper				EMBOSSED (EMB)		

X8L

SIZE		0603	0805		1206		1210	
Soldering		Reflow/Wave	Reflow/Wave		Reflow/Wave		Reflow/Wave	
(L) Length	mm (in.)	1.6 $\pm$ 0.15 (0.063 $\pm$ 0.006)	2.01 $\pm$ 0.2 (0.079 $\pm$ 0.008)		3.2 $\pm$ 0.2 (0.126 $\pm$ 0.008)		3.2 $\pm$ 0.2 (0.126 $\pm$ 0.008)	
(W) Width	mm (in.)	0.81 $\pm$ 0.15 (0.032 $\pm$ 0.006)	1.25 $\pm$ 0.2 (0.049 $\pm$ 0.008)		1.6 $\pm$ 0.2 (0.063 $\pm$ 0.008)		2.5 $\pm$ 0.2 (0.098 $\pm$ 0.008)	
(t) Terminal	mm (in.)	0.35 $\pm$ 0.15 (0.014 $\pm$ 0.006)	0.5 $\pm$ 0.25 (0.02 $\pm$ 0.01)		0.5 $\pm$ 0.25 (0.02 $\pm$ 0.01)		0.5 $\pm$ 0.25 (0.02 $\pm$ 0.01)	
WVDC		25V	50V	100V	25V	50V	100V	16V
271	Cap 270	A	A					
331	(pF) 330	A	A	A	B	B	B	
471	470	A	A	A	B	B	B	
681	680	A	A	A	B	B	B	
102	1000	A	A	A	B	B	B	
152	1500	A	A	A	B	B	B	
182	1800	A	A	A	B	B	B	
222	2200	A	A	A	B	B	B	
272	2700	A	A	A	B	B	B	
332	3300	A	A	A	B	B	B	
392	3900	A	A	A	B	B	B	
472	4700	A	A	A	B	B	B	
562	5600	A	A	A	B	B	B	
682	6800	A	A	A	B	B	B	
822	8200	A	A	A	B	B	B	
103	Cap 0.01	A	A	A	B	B	B	
123	(uF) 0.012	A	A	A	B	B	B	
153	0.015	A	A	A	B	B	B	
183	0.018	A	A	A	B	B	B	
223	0.022	A	A	A	B	B	B	
273	0.027	A	A	A	B	B	B	
333	0.033	A	A	B	B	B	A	
393	0.039	A	A		B	B	A	
473	0.047	A	A		B	B	A	
563	0.056	A	A		B	B	A	
683	0.068	A	A		B	B	A	
823	0.082	A	A		B	B	A	
104	0.1	A	A		B	B	A	
124	0.12				B	A		
154	0.15				B	A		
184	0.18				A	A		
224	0.22				A	A		
274	0.27				A	A		
334	0.33				A	A		
394	0.39				A	A		
474	0.47				A	A		
684	0.68				A	A		
824	0.82				A	A		
105	1				A	A		
155	1.5				A			
225	2.2				A			
475	4.7							
106	10							
WVDC		25V	50V	100V	25V	50V	100V	16V
SIZE		0603		0805		1206		1210

Automotive MLCC - X8G

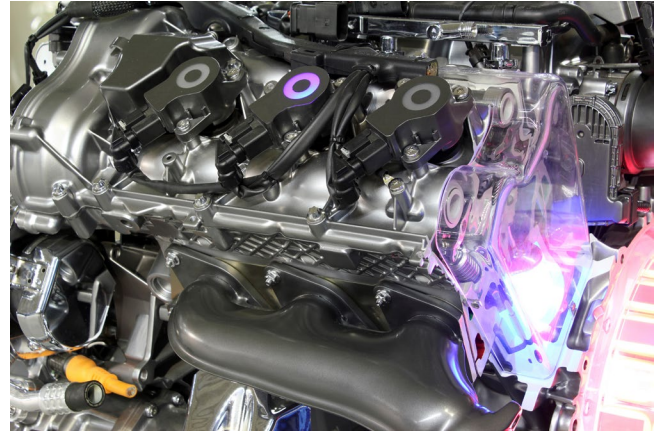
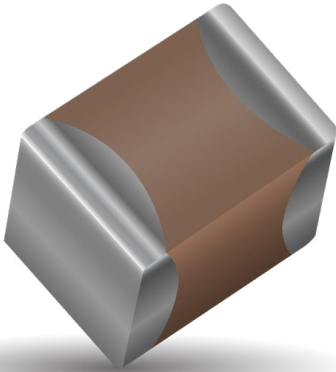
General Specifications

TYPICAL APPLICATIONS

- Extreme capacitance stability
- High temperature
- Battery Management Systems
- Powertrain Sensors & Actuators
- Engine management
- Transmission control
- Safety

ENGINEERING TOOLS

- Samples
- Technical Articles
- Application Engineering
- Application Support



Automotive X8G (-55°C to 150°C , ±30ppm/°C)

Capacitance Range



SIZE		0402		0603		0805	
Soldering		Reflow/Wave		Reflow/Wave		Reflow/Wave	
(L) Length	mm (in.)	1 ± 0.1 (0.04 ± 0.004)		1.6 ± 0.15 (0.063 ± 0.006)		2.01 ± 0.2 (0.079 ± 0.008)	
(W) Width	mm (in.)	0.5 ± 0.1 (0.02 ± 0.004)		0.81 ± 0.15 (0.032 ± 0.006)		1.25 ± 0.2 (0.049 ± 0.008)	
(t) Terminal	mm (in.)	0.25 ± 0.15 (0.01 ± 0.006)		0.35 ± 0.15 (0.014 ± 0.006)		0.5 ± 0.25 (0.02 ± 0.01)	
WVDC		25V	50V	25V	50V	50V	100V
0R5	0.5			A	A	B	B
1R0	1.0			A	A	B	B
1R2	1.2			A	A	B	B
1R5	1.5			A	A	B	B
1R8	1.8			A	A	B	B
2R2	2.2			A	A	B	B
2R7	2.7			A	A	B	B
3R3	3.3			A	A	B	B
3R9	3.9			A	A	B	B
4R7	4.7			A	A	B	B
5R0	5			A	A	B	B
5R6	5.6			A	A	B	B
6R8	6.8			A	A	B	B
8R2	8.2			A	A	B	B
100	10			A	A	B	B
120	12			A	A	B	B
150	15			A	A	B	B
180	18			A	A	B	B
220	22			A	A	B	B
270	27			A	A	B	B
330	33			A	A	B	B
390	39			A	A	B	B
470	47	A	A	A	A	B	B
510	51	A	A	A	A	B	B
560	56	A	A	A	A	B	B
680	68	A	A	A	A	B	B
820	82	A	A	A	A	B	B
101	100	A	A	A	A	B	B
121	120			A	A	B	B
151	150			A	A	B	B
181	180			A	A	B	B
221	220			A	A	B	B
271	270			A	A		
331	330			A	A		
391	390			A	A		
471	470			A	A		
561	560			A	A		
681	680			A	A		
821	820						
102	1000						
122	1200						
152	1500						
182	1800						
222	2200						
272	2700						
332	3300						
392	3900						
472	4700						
562	5600						
682	6800						
103	10nF						
WVDC		25V	50V	25V	50V	50V	100V
Size		0402		0603		0805	

Case Size	0402(KAM05)	0603(KAM15)	0805(KAM21)
Letter	A	A	B
Max Thickness mm	0.56	0.90	0.94
Carrier Tape	Paper	Paper	Paper
Packaging Code 7"reel	H	T	T
Packaging Code 13"reel	N	M	M
	Paper		

Mouser Electronics

Authorized Distributor

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KYOCERA AVX:

KAM15CT70J106KT	KAM05CT70J225KH	KAM21AT70J226KU	KAM03CT70J105KH	KAM05CT70J475KH
KAM15CT70J226KT	KAM32HR73A332KU	KAM32LCG2J223JU	KAM15BR71E224KT	KAM21KR71H225KU
KAM31GCG2J103JU	KAM32LCG2J333JU	KAM32LCG3A103JU	KAM32LL81E106KU	KAM32LL81H475KU
KAM15ACG1E103JT	KAM31GR71C106KU	KAM32GCG3A472JU	KAM32KCG3A102JU	KAM05ACG1H0R5CH
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KAM05ACG1H120JH	KAM05ACG1H150FH	KAM05ACG1H150FN	KAM05ACG1H150JH	KAM05ACG1H180FH
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KAM05ACG1H220JH	KAM05ACG1H270FH	KAM05ACG1H270JH	KAM05ACG1H330FH	KAM05ACG1H330JH
KAM05ACG1H390FH	KAM05ACG1H390FN	KAM05ACG1H390JH	KAM05AR71C103JH	KAM05AR71C103KH
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KAM05AR71C273JH	KAM05AR71C273KH	KAM05AR71C333KH	KAM05AR71C472KH	KAM05AR71C472KN
KAM05AR71C682KH	KAM05AR71C682KN	KAM05AR71E103KH	KAM05AR71E103KN	KAM05AR71H102JH
KAM05AR71H102KH	KAM05AR71H102KN	KAM05AR71H102MH	KAM05AR71H103KH	KAM05AR71H122JH
KAM05AR71H122KH	KAM05AR71H152JH	KAM05AR71H152KH	KAM05AR71H152KN	KAM05AR71H182JH
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KAM05AR71H222JH	KAM05AR71H222JN	KAM05AR71H222KH	KAM05AR71H222KN	KAM05AR71H271JH
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KAM05AR71H332JH	KAM05AR71H332KH	KAM05AR71H332KN	KAM05AR71H391JH	KAM05AR71H391KH
KAM05AR71H392JH	KAM05AR71H392KH	KAM05AR71H471JH	KAM05AR71H471KH	KAM05AR71H471KN
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